Aryan Shakya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.31
Sir Padampat Singhania Education Centre	Class XII (CBSE)	2023	95.40%
Sir Padampat Singhania Education Centre	Class X (CBSE)	2021	97.60%

TECHNICAL & SCHOLASTIC ACHIEVEMENTS

- Secured **1st Place** at the **KLA AI** Hackathon for building a supervised image restoration pipeline.
- Secured **3rd Place** at the **Vivriti Capital** Hackathon for a full-stack AI-powered underwriting platform.
- Recipient of a **scholarship** from State Bank of India for academic excellence.

PROJECTS

Supervised Image Restoration Pipeline for Structured/Textured Images - Github

- Built a supervised grayscale restoration pipeline (NAFNet/NAFSSR default, MPRNet fallback) with a **6-stage analytic feature ablation -** guided filtering, Scharr, DoG, local variance - to preserve fine textures.
- Ran systematic ablations across all 6 feature configs (A0-A5) to isolate each analytic feature's contribution, with the best fold-0 model (NAFSSR-T x2) reaching **28.14 PSNR / 0.7524 SSIM u**nder x8 self-ensemble inference.

Intelli-Credit Copilot - AI-Powered Underwriting Workbench - Github

- Architected a full-stack loan underwriting platform (**FastAPI, async SQLAlchemy, Next.js 14**) covering onboarding, document upload, schema-guided extraction, Five Cs scoring, and automated CAM report generation with export.
- Built a dual-path document processing runtime - **PyMuPDF/pdfplumber** for digital PDFs, **Gemini vision** fallback for scanned pages - with geometry capture and candidate-page retrieval, benchmarked against Docling

Secure Media Vault - Zero-Knowledge E2EE File Sharing Platform - Github

- Built a client-side E2EE platform using the browser's WebCrypto API, deriving keys via PBKDF2-SHA256 **(100K iterations)** and encrypting each chunk with AES-**256-GCM**, with filenames also encrypted to prevent metadata leakage.
- Streamed files in **5MB** chunks via FastAPI + Supabase, achieving **O(1) memory complexity** regardless of file size.

Repo Assistant AI - Github

- Engineered an offline RAG pipeline for codebase Q&A using AST-based chunking, MiniLM embeddings, and Qdrant with disk persistence, achieving **86.67% Hit Rate@5** and **0.74 MRR** on a blind hold-out set .
- Resolved instruction-following limitations by integrating **Phi-3-mini** to process dense AST chunks, achieving **93% correctness** with no hallucinated outputs on the hold-out set.

Decoder-Only Transformer Language Model from Scratch - Github

- Implemented a decoder-only Transformer from scratch in PyTorch (**10.7M params, 6 layers, 6 heads, 384-dim embeddings**), building Multi-Head Attention, FeedForward, and Transformer Block modules natively.
- Trained a character-level GPT on tinyshakespeare for **5,000** iterations, reaching a **1.48 validation loss**

SKILLS

Programming & CS Fundamentals: Python, C/C++, SQL, LaTeX, Data Structures & Algorithms, OOPS, DBMS

Machine Learning: Deep Learning, CNNs, Transformers, RAG, FAISS, PyTorch, TensorFlow, Scikit-learn, LangChain

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

RELEVANT COURSES

CS Courses: Introduction to Programming, Data Structures and Algorithms, Introduction to AI, Computational Physics

Math Courses: Linear Algebra, Probability, Differential Equations, Statistics, Calculus, Advanced Mathematical Physics

Physics Courses : Quantum Mechanics, Optics & Photonics, Analog Electronics, Digital Circuits, Electricity & Magnetism

POSITIONS OF RESPONSIBILITY

Coordinator, Machine Learning, Tinkerers' Lab, IITH

- Taught workshops to 100+ undergrads, covering neural networks and PyTorch through hands-on sessions

Internship Coordinator, Office of Career Services, IITH

- Managed outreach to companies, coordinating partnerships to secure internship opportunities for students

Coordinator, Outreach and Collaborations, TEDxIITHyderabad

- Organized and ran a Model UN event, and led outreach to schools and colleges to drive event partnerships and attendance

EXTRACURRICULAR

- Solved **450+ Data Structures and Algorithms** on **LeetCode**, strengthening problem-solving and optimisation skills
- Volunteered with **National Service Scheme(NSS)** at **IIT Hyderabad**, promoting **community service** through **social outreach.**
- Secured multiple medals in **powerlifting** and **weightlifting** competitions.

CERTIFIED COURSES

- **Machine Learning Specialization** - Stanford & DeepLearning.AI
- **Neural Networks and Deep Learning** - DeepLearning.AI